Supplementary Material for "Neural Forecasters Versus Closed-Form Baselines on Steel-Plant Motor Telemetry: An Open Dataset and a Leakage-Controlled Benchmark"

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This document collects every result table behind the paper, the settings of all models, the data dictionary and three additional figures. All tables are generated by `code/paper/make_paper_tables.py` from the result files in the public repository (`https://github.com/GHOSTCALL983/ahm-motor-telemetry-benchmark`); no number was typed by hand. Notation follows the paper: a *run* is one dataset–target pair (45 runs: 27 vibration, 18 temperature); the operating horizon is 3 min for vibration and 15 min for temperature; "ratio" is a median over runs of an MSE ratio; "wins" counts runs.

A. How the results were verified

Every step from the raw sensor exports to the numbers in the paper was re-implemented and checked by scripts in the repository (`code/audit/`). In summary:

- The data pipeline was rebuilt from the raw exports with anonymised labels; all nine released files are reproduced value by value (maximum difference 2.4×10^{-7}, CSV rounding).
- For all 45 trained runs, the stored ground truth equals the data at every stored origin, every origin lies in its fold's test block, every window lies inside one segment, test blocks are disjoint and no forecast is missing.
- The training code was read for look-ahead: inputs are the 128 rows before the origin, standardisation uses training rows only, early stopping uses a separate validation block, and columns derived from the target are removed.
- The complete analysis was re-run from the released files; all 309 result files (tables, figures, forecasts) were byte-identical to the stored ones.
- An automated final check (Table S25) recomputes the headline numbers independently from the raw forecast files and scans every released file for site or person identifiers.

B. Compute environment

Main neural runs and extra-seed runs: Kaggle, NVIDIA Tesla P100, PyTorch 2.7.1+cu126. Hyperparameter check: Kaggle, NVIDIA Tesla T4, PyTorch 2.11.0+cu130 (18 jobs run from three authors' own accounts, each with a private, checksum-identical copy of the data). The NeuralForecast version was not logged. Training-free methods: CPU; Chronos models via `chronos-forecasting` (`amazon/chronos-bolt-small`, `amazon/chronos-2`).

C. Funding, confidentiality and licences

This work was supported by the Institute of Engineering & Management (IEM), Kolkata, India. The data are released with the permission of the plant; the plant's identity and location, the raw sensor exports, the mapping from device labels to plant assets, and the motor specifications and maintenance records are kept confidential. Everything in the repository is free to use: code under the MIT licence, data and results under Creative Commons Attribution 4.0.

D. Further disclosures

The following points do not change any result but are stated so that readers can judge them.

- The per-device gravity constant uses the device's full record, test periods included. It is a fixed unit conversion that applies equally to all methods; the order of the neighbour roles (nb1, nb2) was likewise fixed from iteration-1 data.
- During neural training, input windows may span a gap between segments (the training script uses a contiguous time axis). Evaluation windows never do.
- The training logs round MSE to four decimals; every number reported here is recomputed from the stored forecasts instead.
- The stored training configurations recorded `iteration = 1` for every run (a metadata bug, corrected in the released configurations); some comments in the as-run training script are out of date. The script itself is released unchanged apart from removed identifiers.
- The variance floor used for the floor-aware training variant (Table S23) was computed on the full record; that variant is therefore not causal and is reported for completeness only.
- An exploratory weight-merging experiment chose its mixing weights on test data. It is not used anywhere in the paper; its file is kept and marked as flagged in the repository.
- A variant of Chronos-2 that also receives the sensor covariates is implemented but was not run to completion; no partial result was kept or reported.
- The 60-minute exponentially weighted mean is restarted at each segment boundary.

E. Figures

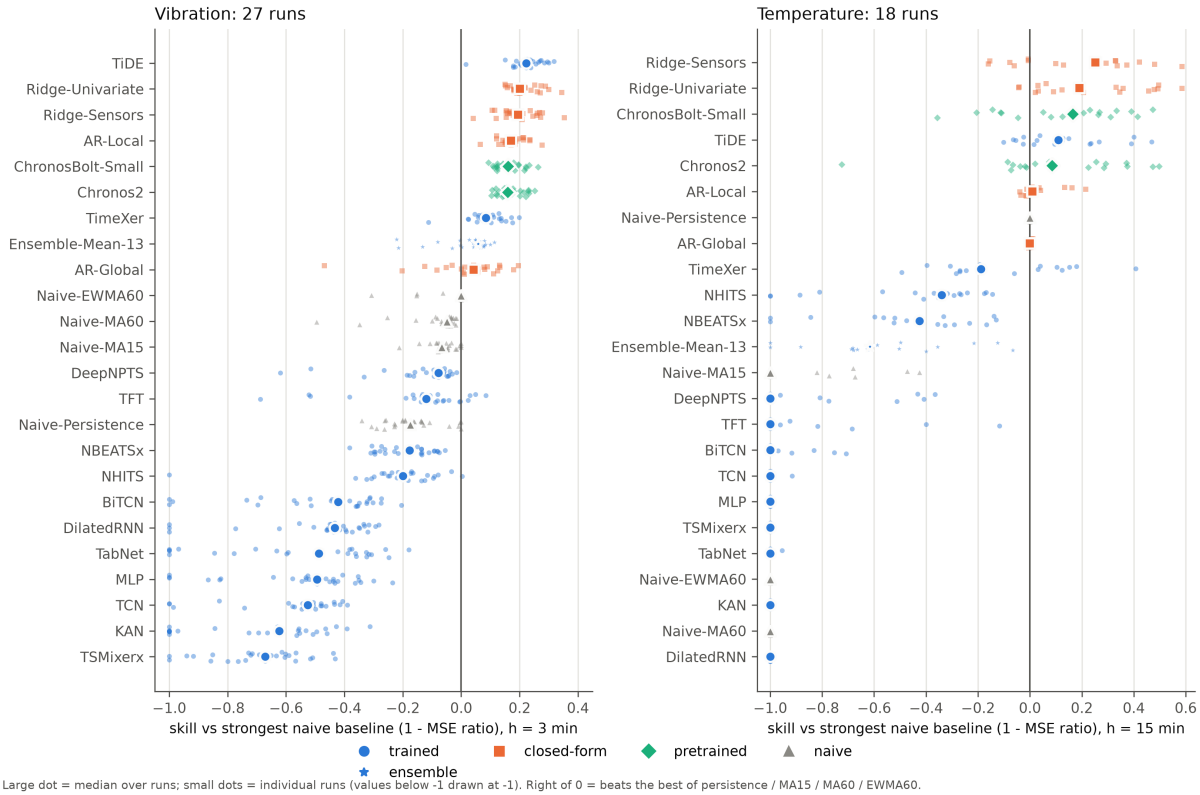


Figure S1: Skill against the strongest naive baseline for every method and run at the operating horizon. Large markers: medians over runs; small markers: individual runs (values below −1 drawn at −1).

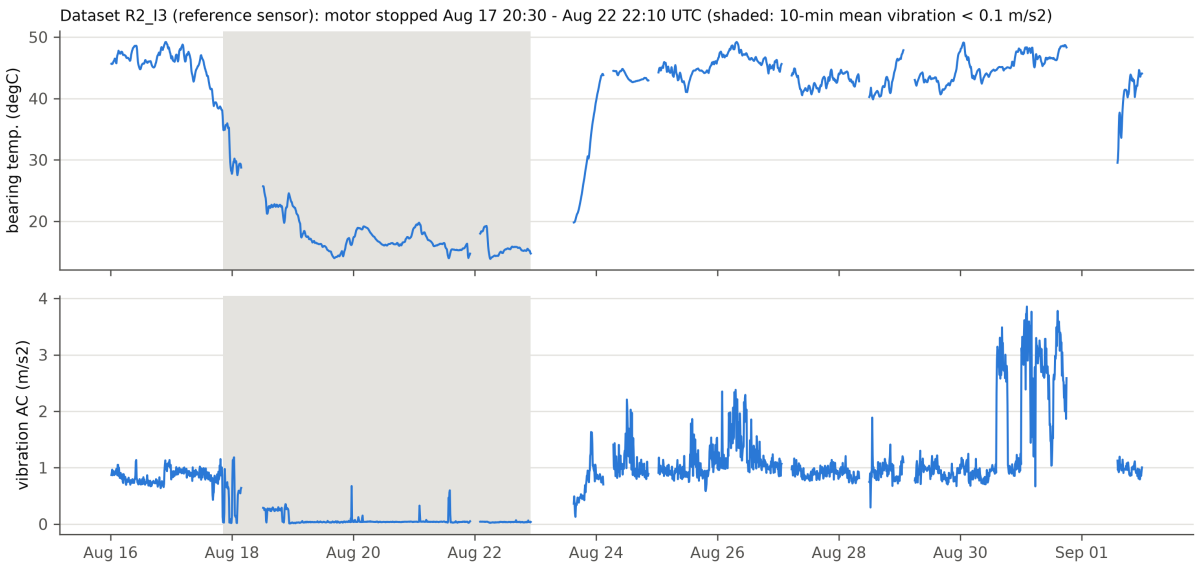


Figure S2: Dataset R2_I3, reference motor: the stop of 17–22 August (shaded) and the high-vibration episode at the end of the record. Both are kept in the data and flagged in `data/anomaly_intervals.csv`.

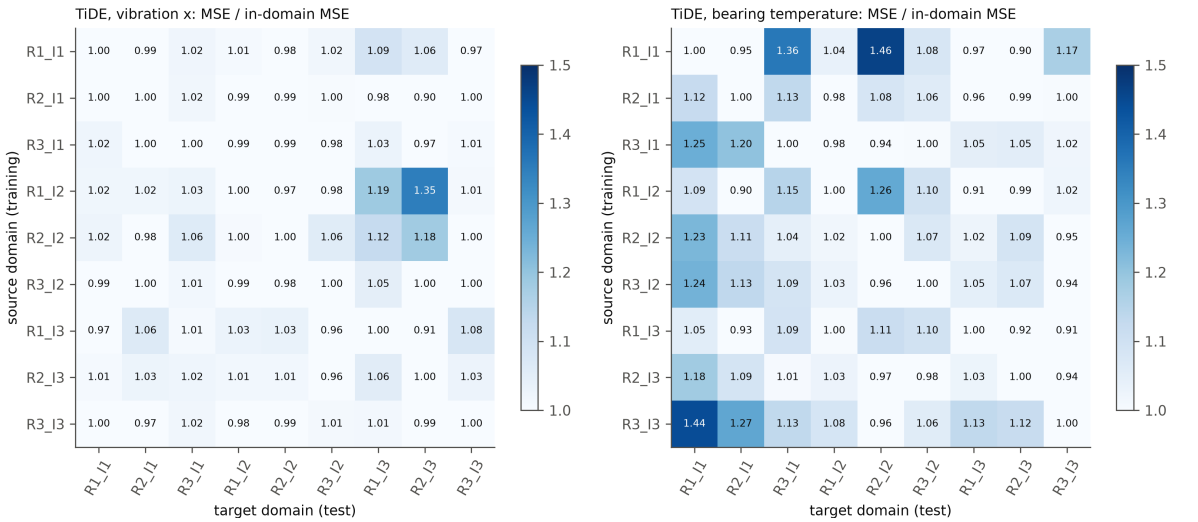


Figure S3: Zero-shot transfer of TiDE between datasets: test MSE on the target dataset divided by the in-domain MSE (fold F5, 80 origins, one seed per cell).

F. Tables

Table S1: Anonymised devices, roles and exclusions.

Device	Region	Zone	Role	Status	Note
AHM-01	R1	Hot	self (reference sensor)	retained	
AHM-02	R1	Hot	nb1	retained	
AHM-03	R1	Hot	nb2	retained	
AHM-04	R2	Mid	nb1	retained	
AHM-05	R2	Mid	nb2	retained	
AHM-06	R2	Mid	self (reference sensor)	retained	
AHM-07	R3	Cool	nb1	retained	one start-up temperature_two sample of 821.9 degC removed by the -20..250 degC envelope
AHM-08	R3	Cool	self (reference sensor)	retained	
AHM-09	R3	Cool	-	excluded	not faulty; uptime misaligned with AHM-07/AHM-08 so three-way overlap would roughly halve usable minutes
AHM-10	-	-	-	excluded	starts after the windows open; 21.7% coverage of its own span; y axis reads exactly 0.0 in 50.98% of samples (x and z also mostly constant); temperature spike 835.51 degC
AHM-11	-	-	-	excluded	1.0% coverage (21728 rows); temperature spike 833.51 degC

Table S2: Raw exports per device: sample count, mean samples per active minute, coverage of the device's own span, and the gravity constant k_d (median static resultant in raw units) with its inter-quartile range in percent of k_d.

Device	Raw samples	Samples/min	Coverage (%)	k_d	IQR (%)
AHM-01	1,446,228	14.0	87.2	1.8479	3.65
AHM-02	3,897,155	14.2	66.7	1.8388	2.69
AHM-03	1,435,852	15.6	35.6	1.8266	7.63
AHM-04	1,921,174	14.8	32.7	1.8275	4.26
AHM-05	3,858,958	12.7	55.7	1.8120	2.15
AHM-06	1,999,901	15.7	36.8	1.7904	5.07
AHM-07	2,180,603	15.0	53.4	1.7935	4.17
AHM-08	1,833,745	15.3	42.0	1.8065	7.08
AHM-09	1,957,797	16.3	29.5	1.8244	2.53
AHM-10	1,022,216	21.0	21.7	1.8068	0.55
AHM-11	21,728	21.5	1.0	0.1905	10.98

Table S3: Released files. Segment lengths in minutes; origin capacity counts every admissible forecast origin (the evaluation uses 400 per run); y is the reference-motor bearing temperature in °C.

Dataset	Rows	Seg.	Min seg.	Max seg.	Feat.	Origin cap.	$y_{\min}$	$y_{\max}$
R1_I1	16,463	10	314	4,158	56	15,013	31.7	73.7
R2_I1	15,983	12	314	3,623	56	14,243	26.5	47.5
R3_I1	22,807	9	911	4,390	56	21,502	23.1	38.4
R1_I2	16,722	18	257	3,835	56	14,112	34.2	80.3
R2_I2	15,938	18	227	3,835	56	13,328	28.7	50.2
R3_I2	21,485	8	364	6,411	56	20,325	21.9	37.0
R1_I3	13,684	8	248	3,614	56	12,524	31.5	83.1
R2_I3	20,238	10	582	4,911	56	18,788	13.9	49.3
R3_I3	14,387	3	753	11,676	56	13,952	20.2	35.3

Table S4: All run-level paired comparisons at the operating horizon (median ratio of MSE, method over reference; Wilcoxon signed-rank; Holm within kind).

Kind	Method	Reference	Ratio	Wins	p	p_{Holm}
temperature	Ridge (univariate)	TiDE	0.872	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Ridge (+sensors)	TiDE	0.910	13/18	0.02	0.05
temperature	Chronos-2	TiDE	0.954	11/18	0.37	0.74
temperature	Chronos-Bolt-S	TiDE	0.973	11/18	0.80	0.80
temperature	AR-Local	TiDE	1.095	5/18	9.0×10^{-3}	0.05
temperature	AR-Global	TiDE	1.123	5/18	0.01	0.05

Kind	Method	Reference	Ratio	Wins	p	p_{Holm}
temperature	Persistence	TiDE	1.124	5/18	7.7×10^{-3}	0.05
temperature	Ensemble mean (13)	TiDE	1.790	0/18	7.6×10^{-6}	2.1×10^{-4}
temperature	MA-15	TiDE	2.410	0/18	7.6×10^{-6}	2.1×10^{-4}
temperature	EWMA-60	TiDE	5.116	0/18	7.6×10^{-6}	2.1×10^{-4}
temperature	MA-60	TiDE	6.664	0/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Ridge (univariate)	median trained model	0.262	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Ridge (+sensors)	median trained model	0.275	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Chronos-2	median trained model	0.281	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Chronos-Bolt-S	median trained model	0.299	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	AR-Local	median trained model	0.330	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Persistence	median trained model	0.341	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	AR-Global	median trained model	0.342	18/18	7.6×10^{-6}	2.1×10^{-4}
temperature	Ensemble mean (13)	median trained model	0.549	17/18	1.5×10^{-5}	2.3×10^{-4}
temperature	MA-15	median trained model	0.757	15/18	1.3×10^{-3}	0.01
temperature	EWMA-60	median trained model	1.562	2/18	8.4×10^{-4}	7.6×10^{-3}
temperature	MA-60	median trained model	2.006	2/18	5.3×10^{-5}	6.9×10^{-4}
vibration	Ridge (univariate)	TiDE	1.024	6/27	1.3×10^{-3}	0.01
vibration	Ridge (+sensors)	TiDE	1.048	2/27	1.6×10^{-4}	1.6×10^{-3}
vibration	AR-Local	TiDE	1.068	1/27	1.9×10^{-5}	2.6×10^{-4}
vibration	Chronos-2	TiDE	1.093	4/27	5.5×10^{-5}	6.9×10^{-4}
vibration	Chronos-Bolt-S	TiDE	1.094	3/27	6.3×10^{-5}	7.0×10^{-4}
vibration	Ensemble mean (13)	TiDE	1.240	0/27	1.5×10^{-8}	6.6×10^{-7}
vibration	AR-Global	TiDE	1.251	1/27	3.0×10^{-8}	8.6×10^{-7}
vibration	EWMA-60	TiDE	1.311	0/27	1.5×10^{-8}	6.6×10^{-7}
vibration	MA-60	TiDE	1.356	0/27	1.5×10^{-8}	6.6×10^{-7}
vibration	MA-15	TiDE	1.383	0/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Persistence	TiDE	1.528	0/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Ridge (univariate)	median trained model	0.580	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Ridge (+sensors)	median trained model	0.600	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Chronos-2	median trained model	0.616	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	AR-Local	median trained model	0.622	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Chronos-Bolt-S	median trained model	0.624	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	AR-Global	median trained model	0.698	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Ensemble mean (13)	median trained model	0.704	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	EWMA-60	median trained model	0.742	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	MA-60	median trained model	0.783	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	MA-15	median trained model	0.792	27/27	1.5×10^{-8}	6.6×10^{-7}
vibration	Persistence	median trained model	0.869	24/27	2.1×10^{-7}	5.8×10^{-6}

Table S5: Median R^2 over runs (fold-averaged) and mean skill at the operating horizon. Mean skill is pulled down by diverged forecasts; medians are reported in the paper.

Method	Vib. median R^2	Vib. mean skill	Temp. median R^2	Temp. mean skill
TiDE	0.379	+0.224	0.963	+0.128
Ridge (univariate)	0.357	+0.216	0.967	+0.234
Ridge (+sensors)	0.350	+0.195	0.964	+0.190
AR-Local	0.342	+0.173	0.956	+0.031
Chronos-2	0.322	+0.169	0.964	+0.108
Chronos-Bolt-S	0.317	+0.163	0.963	+0.121
TimeXer	0.285	+0.089	0.955	-0.104
Ensemble mean (13)	0.272	+0.021	0.929	-0.680
AR-Global	0.237	+0.021	0.955	-0.000
EWMA-60	0.223	-0.025	0.823	-3.803
MA-60	0.181	-0.086	0.781	-5.017
MA-15	0.170	-0.062	0.909	-1.134
DeepNPTS	0.140	-0.137	0.911	-1.368
TFT	0.139	-0.143	0.891	-1.374
NBEATSx	0.075	-0.194	0.941	-0.681
NHITS	0.068	-0.226	0.932	-8.380
Persistence	0.036	-0.168	0.955	+0.000
BiTCN	-0.103	-0.541	0.877	-1.815
DilatedRNN	-0.107	-0.637	0.720	-4.906
TabNet	-0.147	-0.758	0.811	-4.417
MLP	-0.162	-0.598	0.831	-2.720
KAN	-0.207	-0.793	0.774	-7.558
TCN	-0.210	-0.709	0.838	-2.436
TSMixerx	-0.302	-0.705	0.850	-2.859

Table S6: Skill against the strongest naive baseline for every run at the operating horizon (vibration $h = 3$, temperature $h = 15$).

Run	TiDE	TimeXer	Ridge-U	Ridge-S	Bolt-S	Chr.-2	Med. trained	Best naive
R1_I1 temperature_one	+0.266	+0.104	+0.479	+0.329	+0.390	+0.375	-2.226	Persistence
R1_I1 temperature_two	+0.026	+0.040	+0.032	-0.155	-0.110	+0.073	-1.077	Persistence
R1_I1 x_ac	+0.245	+0.107	+0.198	+0.154	+0.148	+0.148	-0.383	EWMA-60
R1_I1 y_ac	+0.214	+0.085	+0.210	+0.212	+0.094	+0.106	-0.307	EWMA-60
R1_I1 z_ac	+0.222	+0.069	+0.201	+0.167	+0.124	+0.120	-0.349	EWMA-60
R1_I2 temperature_one	+0.234	-0.246	+0.346	+0.336	+0.267	+0.285	-2.442	Persistence
R1_I2 temperature_two	-0.064	-0.284	-0.042	-0.060	-0.356	-0.074	-2.793	Persistence
R1_I2 x_ac	+0.185	+0.024	+0.158	+0.119	+0.184	+0.202	-0.307	EWMA-60
R1_I2 y_ac	+0.211	+0.076	+0.181	+0.159	+0.176	+0.154	-0.313	EWMA-60
R1_I2 z_ac	+0.151	+0.024	+0.150	+0.106	+0.196	+0.165	-0.308	EWMA-60
R1_I3 temperature_one	+0.125	-0.186	+0.359	+0.330	+0.207	+0.225	-1.917	Persistence
R1_I3 temperature_two	-0.025	-0.494	+0.062	-0.006	-0.144	+0.019	-2.497	Persistence
R1_I3 x_ac	+0.214	+0.130	+0.168	+0.195	+0.123	+0.156	-0.520	MA-15
R1_I3 y_ac	+0.178	+0.099	+0.151	+0.041	+0.125	+0.130	-0.324	EWMA-60
R1_I3 z_ac	+0.188	+0.073	+0.190	+0.154	+0.221	+0.213	-0.736	Persistence
R2_I1 temperature_one	+0.390	+0.152	+0.496	+0.373	+0.264	+0.371	-1.931	Persistence
R2_I1 temperature_two	+0.037	-0.269	+0.176	+0.129	-0.047	+0.096	-2.213	Persistence
R2_I1 x_ac	+0.230	+0.102	+0.190	+0.173	+0.159	+0.147	-0.205	EWMA-60
R2_I1 y_ac	+0.256	+0.085	+0.219	+0.199	+0.166	+0.174	-0.210	EWMA-60
R2_I1 z_ac	+0.217	+0.058	+0.201	+0.196	+0.150	+0.168	-0.314	EWMA-60
R2_I2 temperature_one	+0.126	+0.031	+0.206	+0.264	+0.231	-0.013	-1.529	Persistence
R2_I2 temperature_two	-0.055	-0.192	-0.043	-0.078	-0.205	-0.085	-0.962	Persistence
R2_I2 x_ac	+0.179	+0.054	+0.179	+0.115	+0.112	+0.118	-0.348	EWMA-60
R2_I2 y_ac	+0.201	+0.129	+0.181	+0.154	+0.161	+0.177	-0.480	EWMA-60
R2_I2 z_ac	+0.205	+0.071	+0.193	+0.168	+0.145	+0.159	-0.449	EWMA-60
R2_I3 temperature_one	+0.149	+0.179	+0.332	+0.294	+0.311	+0.254	-1.345	Persistence
R2_I3 temperature_two	-0.101	-0.307	+0.021	-0.160	-0.114	-0.019	-1.289	Persistence
R2_I3 x_ac	+0.016	-0.112	+0.289	+0.278	+0.233	+0.221	-1.016	Persistence
R2_I3 y_ac	+0.300	+0.198	+0.300	+0.273	+0.264	+0.230	-0.357	MA-15
R2_I3 z_ac	+0.318	+0.111	+0.345	+0.353	+0.161	+0.252	-1.242	MA-15
R3_I1 temperature_one	+0.469	+0.407	+0.586	+0.586	+0.473	+0.473	-0.917	Persistence
R3_I1 temperature_two	+0.095	-0.195	+0.123	+0.239	+0.125	-0.066	-0.998	Persistence
R3_I1 x_ac	+0.265	+0.175	+0.240	+0.236	+0.196	+0.180	-0.273	EWMA-60
R3_I1 y_ac	+0.269	+0.138	+0.252	+0.245	+0.180	+0.202	-0.303	EWMA-60
R3_I1 z_ac	+0.238	+0.153	+0.204	+0.191	+0.122	+0.122	-0.382	EWMA-60
R3_I2 temperature_one	+0.231	-0.060	+0.457	+0.479	+0.333	+0.292	-2.308	Persistence
R3_I2 temperature_two	-0.024	-0.397	+0.030	+0.100	+0.051	-0.725	-2.051	Persistence
R3_I2 x_ac	+0.295	+0.137	+0.261	+0.261	+0.216	+0.230	-0.324	EWMA-60
R3_I2 y_ac	+0.285	+0.172	+0.271	+0.259	+0.163	+0.151	-0.423	EWMA-60
R3_I2 z_ac	+0.237	+0.075	+0.203	+0.208	+0.122	+0.141	-0.300	EWMA-60
R3_I3 temperature_one	+0.400	+0.124	+0.493	+0.422	+0.414	+0.497	-1.937	Persistence
R3_I3 temperature_two	+0.015	-0.270	+0.091	-0.007	+0.082	-0.037	-2.771	Persistence
R3_I3 x_ac	+0.225	+0.046	+0.218	+0.207	+0.143	+0.148	-0.389	EWMA-60
R3_I3 y_ac	+0.222	+0.060	+0.202	+0.178	+0.108	+0.104	-0.322	EWMA-60
R3_I3 z_ac	+0.273	+0.070	+0.274	+0.268	+0.221	+0.231	-0.461	EWMA-60

Table S7: Median skill by horizon, vibration (cumulative MSE over steps 1..h).

h	TiDE	Ridge-U	Bolt-S	Chronos-2	Persistence	Med. trained
1	+0.269	+0.199	+0.122	+0.125	+0.000	-0.995
2	+0.299	+0.253	+0.201	+0.210	+0.000	-0.455
3	+0.222	+0.202	+0.161	+0.159	-0.174	-0.348
4	+0.160	+0.153	+0.105	+0.109	-0.304	-0.366
5	+0.128	+0.122	+0.080	+0.081	-0.395	-0.360
6	+0.105	+0.105	+0.054	+0.055	-0.454	-0.345
7	+0.088	+0.088	+0.038	+0.042	-0.492	-0.347
8	+0.071	+0.078	+0.027	+0.026	-0.504	-0.369
9	+0.065	+0.069	+0.020	+0.021	-0.522	-0.379
10	+0.056	+0.062	+0.009	+0.012	-0.532	-0.371
11	+0.046	+0.053	+0.001	+0.010	-0.549	-0.371
12	+0.039	+0.049	-0.005	+0.007	-0.559	-0.366
13	+0.033	+0.044	-0.013	-0.000	-0.567	-0.365
14	+0.028	+0.040	-0.019	-0.006	-0.582	-0.363
15	+0.026	+0.036	-0.025	-0.009	-0.598	-0.360

Table S8: Median skill by horizon, temperature (cumulative MSE over steps 1..h).

h	TiDE	Ridge-U	Bolt-S	Chronos-2	Persistence	Med. trained
1	+0.205	+0.426	-0.302	-0.170	+0.000	-58.501
2	+0.193	+0.471	+0.092	+0.134	+0.000	-29.376
3	+0.205	+0.472	+0.232	+0.194	+0.000	-16.738
4	+0.200	+0.432	+0.253	+0.185	+0.000	-11.382
5	+0.191	+0.399	+0.277	+0.168	+0.000	-8.117
6	+0.185	+0.368	+0.269	+0.156	+0.000	-6.113
7	+0.173	+0.341	+0.259	+0.142	+0.000	-4.952
8	+0.160	+0.318	+0.251	+0.132	+0.000	-4.184
9	+0.148	+0.297	+0.239	+0.124	+0.000	-3.742
10	+0.138	+0.279	+0.224	+0.116	+0.000	-3.361
11	+0.131	+0.262	+0.210	+0.109	+0.000	-3.058
12	+0.126	+0.244	+0.197	+0.102	+0.000	-2.735
13	+0.121	+0.226	+0.185	+0.096	+0.000	-2.441
14	+0.114	+0.208	+0.174	+0.090	+0.000	-2.183
15	+0.110	+0.191	+0.166	+0.084	+0.000	-1.934

Table S9: Murphy decomposition of MSE (median over runs). ρ/r is the ratio of forecast to observed standard deviation divided by their correlation; 1 means no conditional bias, > 1 means the forecast swings too much for its correlation.

Kind	Method	ρ/r	r	Bias (%)	Cond. bias (%)	Unexpl. (%)
temperature	Ensemble mean (13)	1.000	0.964	1.5	1.2	97.2
temperature	Ridge (+sensors)	1.004	0.983	2.8	1.3	94.9
temperature	AR-Global	1.007	0.978	0.7	1.2	98.1
temperature	TimeXer	1.008	0.978	2.2	1.8	96.2
temperature	Persistence	1.017	0.978	0.4	1.3	98.1
temperature	Chronos-2	1.018	0.982	0.6	1.5	97.8
temperature	Ridge (univariate)	1.019	0.984	0.3	1.9	97.4
temperature	Chronos-Bolt-S	1.019	0.982	0.7	1.9	97.2
temperature	DeepNPTS	1.020	0.955	1.7	1.8	96.4
temperature	NBEATSx	1.022	0.970	1.0	2.0	97.2
temperature	TiDE	1.023	0.982	0.4	2.6	97.1
temperature	EWMA-60	1.025	0.913	1.7	1.5	96.9
temperature	AR-Local	1.025	0.978	0.3	2.4	97.3
temperature	TFT	1.029	0.947	3.1	2.2	94.1
temperature	NHITS	1.030	0.966	0.9	2.2	97.0
temperature	MA-15	1.031	0.954	0.7	2.1	97.1
temperature	TabNet	1.053	0.910	10.3	5.7	81.9
temperature	BiTCN	1.054	0.942	1.7	5.8	92.5
temperature	MA-60	1.072	0.890	1.4	3.2	94.9
temperature	TCN	1.093	0.920	1.5	6.0	90.6
temperature	TSMixerx	1.098	0.929	2.0	6.4	90.9
temperature	MLP	1.098	0.923	1.9	7.1	91.2
temperature	DilatedRNN	1.124	0.864	2.2	8.5	87.8
temperature	KAN	1.130	0.902	2.3	12.0	84.7
vibration	Ensemble mean (13)	1.006	0.514	0.9	1.3	98.0
vibration	Ridge (univariate)	1.077	0.602	0.9	0.9	98.0
vibration	Ridge (+sensors)	1.093	0.588	1.2	1.0	97.5
vibration	AR-Local	1.100	0.583	0.9	1.1	98.0
vibration	TiDE	1.113	0.619	1.2	1.1	97.7
vibration	AR-Global	1.187	0.541	4.6	2.9	90.8
vibration	EWMA-60	1.192	0.469	0.7	1.0	98.4
vibration	Chronos-Bolt-S	1.253	0.588	0.8	3.1	96.0
vibration	Chronos-2	1.259	0.593	0.8	3.8	95.7
vibration	MA-60	1.332	0.449	0.6	1.8	97.6
vibration	TimeXer	1.337	0.552	0.9	3.5	95.8
vibration	TabNet	1.341	0.128	8.3	6.8	81.4
vibration	TFT	1.520	0.437	1.4	5.7	93.5
vibration	MA-15	1.683	0.453	0.6	5.4	94.0
vibration	DeepNPTS	1.699	0.398	0.8	5.2	93.9
vibration	NHITS	1.838	0.450	1.1	13.8	85.5
vibration	Persistence	1.974	0.514	0.9	22.6	76.3
vibration	NBEATSx	1.982	0.448	1.1	15.1	83.7
vibration	BiTCN	2.510	0.309	1.7	15.0	83.3
vibration	DilatedRNN	3.028	0.289	0.8	16.4	82.8
vibration	KAN	3.055	0.267	0.9	20.4	78.2
vibration	MLP	3.194	0.288	1.1	20.1	79.6
vibration	TCN	3.417	0.245	1.0	21.4	77.6
vibration	TSMixerx	3.567	0.227	1.1	26.2	72.8

Table S10: Causal recalibration by method family (cells = run × method; folds 2–5).

Kind	Family	Variant	Cells	Median ratio	Improved (%)
temperature	closed-form	affine	72	1.013	25.0
temperature	closed-form	anchored	72	1.006	30.6
temperature	closed-form	shrink	72	1.007	25.0
temperature	ensemble	affine	18	1.033	16.7
temperature	ensemble	anchored	18	0.678	100.0
temperature	ensemble	shrink	18	1.012	11.1
temperature	pretrained	affine	36	1.013	25.0
temperature	pretrained	anchored	36	0.973	66.7
temperature	pretrained	shrink	36	1.011	19.4
temperature	trained	affine	234	1.034	20.1
temperature	trained	anchored	234	0.414	94.9
temperature	trained	shrink	234	1.030	18.8
vibration	closed-form	affine	108	1.030	10.2
vibration	closed-form	anchored	108	1.012	16.7
vibration	closed-form	shrink	108	1.020	15.7
vibration	ensemble	affine	27	1.019	22.2
vibration	ensemble	anchored	27	0.863	100.0
vibration	ensemble	shrink	27	1.017	25.9
vibration	pretrained	affine	54	1.020	38.9
vibration	pretrained	anchored	54	1.003	42.6
vibration	pretrained	shrink	54	1.015	42.6
vibration	trained	affine	351	0.972	61.5
vibration	trained	anchored	351	0.694	94.3
vibration	trained	shrink	351	0.974	61.3

Table S11: Anchored recalibration per trained model (MSE after / before, folds 2–5).

Model	Vib. ratio	Improved	Temp. ratio	Improved
KAN	0.559	27/27	0.191	18/18
TSMixerx	0.563	27/27	0.280	18/18
TCN	0.605	27/27	0.321	18/18
MLP	0.606	27/27	0.301	18/18
DilatedRNN	0.641	27/27	0.186	18/18
TabNet	0.651	27/27	0.281	18/18
BiTCN	0.660	27/27	0.433	18/18
NHITS	0.791	27/27	0.760	18/18
DeepNPTS	0.805	27/27	0.468	18/18
TFT	0.839	27/27	0.451	18/18
NBEATSx	0.840	27/27	0.749	18/18
TimeXer	0.935	27/27	0.859	17/18
TiDE	1.008	7/27	1.006	7/18

Table S12: Does recalibration change which trained model is best in a run?

Kind	Variant	Best trained model unchanged
temperature	affine	17/18
temperature	anchored	11/18
temperature	shrink	18/18
vibration	affine	27/27
vibration	anchored	25/27
vibration	shrink	27/27

Table S13: Predictability scores computed on training rows only versus model outcomes. p_{Holm} is adjusted over the four outcomes of each scope and score.

Scope	Score	Outcome	n	Spearman	p	p_{Holm}
all	lambda	share_trained_beat_naive	45	-0.525	2.1×10^{-4}	4.2×10^{-4}
all	lambda	median_overconf	45	-0.781	2.5×10^{-10}	7.6×10^{-10}
all	lambda	median_anchor_ratio	45	-0.786	1.7×10^{-10}	6.6×10^{-10}
all	lambda	best_skill	45	+0.114	0.46	0.46
all	acf_h	share_trained_beat_naive	45	-0.600	1.3×10^{-5}	2.6×10^{-5}
all	acf_h	median_overconf	45	-0.767	7.7×10^{-10}	2.3×10^{-9}

Scope	Score	Outcome	n	Spearman	p	p_{Holm}
all	acf_h	median_anchor_ratio	45	-0.804	2.9×10^{-11}	1.1×10^{-10}
all	acf_h	best_skill	45	+0.132	0.39	0.39
temperature	lambda	share_trained_beat_naive	18	+0.273	0.27	1.00
temperature	lambda	median_overconf	18	-0.022	0.93	1.00
temperature	lambda	median_anchor_ratio	18	-0.038	0.88	1.00
temperature	lambda	best_skill	18	+0.094	0.71	1.00
temperature	acf_h	share_trained_beat_naive	18	-0.237	0.34	1.00
temperature	acf_h	median_overconf	18	-0.022	0.93	1.00
temperature	acf_h	median_anchor_ratio	18	-0.024	0.93	1.00
temperature	acf_h	best_skill	18	-0.067	0.79	1.00
vibration	lambda	share_trained_beat_naive	27	-0.003	0.99	0.99
vibration	lambda	median_overconf	27	-0.274	0.17	0.33
vibration	lambda	median_anchor_ratio	27	-0.416	0.03	0.12
vibration	lambda	best_skill	27	+0.397	0.04	0.12
vibration	acf_h	share_trained_beat_naive	27	-0.111	0.58	0.58
vibration	acf_h	median_overconf	27	-0.256	0.20	0.40
vibration	acf_h	median_anchor_ratio	27	-0.503	7.5×10^{-3}	0.03
vibration	acf_h	best_skill	27	+0.349	0.07	0.22

Table S14: Main comparisons after dropping every origin whose input or forecast window touches a flagged anomaly (17,315 of 18,000 origins kept).

Kind	Comparison	Value	Wins	p
temperature	trained models beating best naive in majority of runs	1.000		–
temperature	Ensemble-Mean-13 vs TiDE: median MSE ratio	1.879	0	7.6×10^{-6}
temperature	Ensemble-Mean-13 vs median trained model: median MSE ratio	0.550	17	1.5×10^{-5}
temperature	Ridge-Univariate vs TiDE: median MSE ratio	0.888	18	7.6×10^{-6}
temperature	Ridge-Univariate vs median trained model: median MSE ratio	0.264	18	7.6×10^{-6}
temperature	Ridge-Sensors vs TiDE: median MSE ratio	0.910	13	0.02
temperature	Ridge-Sensors vs median trained model: median MSE ratio	0.277	18	7.6×10^{-6}
temperature	ChronosBolt-Small vs TiDE: median MSE ratio	0.983	11	0.93
temperature	ChronosBolt-Small vs median trained model: median MSE ratio	0.307	18	7.6×10^{-6}
temperature	Chronos2 vs TiDE: median MSE ratio	0.998	9	0.83
temperature	Chronos2 vs median trained model: median MSE ratio	0.283	18	7.6×10^{-6}
temperature	anchored recalibration of trained models: median MSE ratio	0.403	18	7.6×10^{-6}
vibration	trained models beating best naive in majority of runs	2.000		–
vibration	Ensemble-Mean-13 vs TiDE: median MSE ratio	1.212	1	4.5×10^{-8}
vibration	Ensemble-Mean-13 vs median trained model: median MSE ratio	0.710	27	1.5×10^{-8}
vibration	Ridge-Univariate vs TiDE: median MSE ratio	1.024	5	8.3×10^{-4}
vibration	Ridge-Univariate vs median trained model: median MSE ratio	0.603	27	1.5×10^{-8}
vibration	Ridge-Sensors vs TiDE: median MSE ratio	1.048	2	8.4×10^{-5}
vibration	Ridge-Sensors vs median trained model: median MSE ratio	0.612	27	1.5×10^{-8}
vibration	ChronosBolt-Small vs TiDE: median MSE ratio	1.097	3	4.7×10^{-5}
vibration	ChronosBolt-Small vs median trained model: median MSE ratio	0.649	27	1.5×10^{-8}
vibration	Chronos2 vs TiDE: median MSE ratio	1.091	3	3.0×10^{-5}
vibration	Chronos2 vs median trained model: median MSE ratio	0.652	27	1.5×10^{-8}
vibration	anchored recalibration of trained models: median MSE ratio	0.746	27	1.5×10^{-8}

Table S15: Trained-model cells whose operating-horizon MSE exceeds ten times that of persistence on the same fold (41 of 2,925).

Run	Fold	Model	MSE / persistence MSE
R3_I2 temperature_one	F2	NHITS	526.1
R3_I2 temperature_one	F2	KAN	162.6
R3_I2 temperature_two	F1	TabNet	43.2
R2_I3 temperature_one	F1	TabNet	34.3
R3_I2 temperature_two	F2	KAN	28.3
R3_I3 temperature_one	F1	TabNet	25.9
R2_I3 temperature_two	F3	TabNet	23.8
R1_I1 temperature_two	F4	KAN	23.0
R3_I3 temperature_one	F3	KAN	21.7
R1_I1 temperature_two	F4	TabNet	21.7
R1_I1 temperature_one	F4	KAN	19.5
R2_I2 temperature_two	F2	TabNet	18.6
R1_I2 temperature_two	F5	KAN	18.6
R1_I2 temperature_one	F1	DilatedRNN	18.0
R1_I2 temperature_two	F5	TabNet	17.2
R2_I3 temperature_two	F1	TabNet	16.8

Run	Fold	Model	MSE / persistence MSE
R1_I3 temperature_one	F5	TabNet	15.6
R1_I3 temperature_two	F1	TabNet	15.3
R2_I1 temperature_one	F3	DilatedRNN	13.8
R1_I1 temperature_one	F1	DilatedRNN	13.7
R3_I1 temperature_two	F1	MLP	13.4
R1_I3 temperature_two	F5	BiTCN	13.4
R2_I3 temperature_one	F4	TabNet	12.9
R3_I2 temperature_one	F2	DilatedRNN	12.9
R2_I3 z_ac	F1	BiTCN	12.7
R2_I3 z_ac	F2	TCN	12.6
R2_I2 temperature_one	F1	DilatedRNN	12.6
R3_I1 temperature_two	F1	KAN	12.4
R2_I3 z_ac	F2	DilatedRNN	12.2
R1_I2 temperature_two	F1	TCN	12.1
R3_I2 temperature_one	F1	TabNet	11.9
R2_I2 temperature_two	F1	TabNet	11.9
R1_I2 temperature_two	F1	DilatedRNN	11.7
R3_I2 temperature_two	F3	DilatedRNN	11.5
R3_I1 temperature_two	F1	TCN	11.5
R2_I2 temperature_one	F5	TabNet	11.1
R3_I1 temperature_two	F3	KAN	10.5
R3_I3 temperature_two	F3	DilatedRNN	10.4
R1_I3 temperature_one	F1	DilatedRNN	10.4
R3_I2 temperature_one	F3	DilatedRNN	10.1
R1_I2 temperature_one	F3	KAN	10.1

Table S16: Every headline comparison with and without the diverged cells.

Kind	Claim	As reported	Diverged removed
vibration	trained models beating best naive in a majority of runs	2.000	2.000
vibration	Ridge-Univariate vs TiDE: median MSE ratio	1.024 (6)	1.024 (6)
vibration	Ridge-Univariate vs median trained model: median MSE ratio	0.580 (27)	0.580 (27)
vibration	Ridge-Sensors vs TiDE: median MSE ratio	1.048 (2)	1.048 (2)
vibration	Ridge-Sensors vs median trained model: median MSE ratio	0.600 (27)	0.600 (27)
vibration	AR-Global vs TiDE: median MSE ratio	1.251 (1)	1.251 (1)
vibration	AR-Global vs median trained model: median MSE ratio	0.698 (27)	0.698 (27)
vibration	AR-Local vs TiDE: median MSE ratio	1.068 (1)	1.068 (1)
vibration	AR-Local vs median trained model: median MSE ratio	0.622 (27)	0.622 (27)
vibration	ChronosBolt-Small vs TiDE: median MSE ratio	1.094 (3)	1.094 (3)
vibration	ChronosBolt-Small vs median trained model: median MSE ratio	0.624 (27)	0.624 (27)
vibration	Chronos2 vs TiDE: median MSE ratio	1.093 (4)	1.093 (4)
vibration	Chronos2 vs median trained model: median MSE ratio	0.616 (27)	0.616 (27)
vibration	Ensemble-Mean-13 vs TiDE: median MSE ratio	1.240 (0)	1.240 (0)
vibration	Ensemble-Mean-13 vs median trained model: median MSE ratio	0.704 (27)	0.704 (27)
vibration	Ensemble-Median-13 vs TiDE: median MSE ratio	1.243 (0)	1.243 (0)
vibration	Ensemble-Median-13 vs median trained model: median MSE ratio	0.708 (27)	0.708 (27)
vibration	best naive (reference)	0.004	0.004
vibration	recalibration 'shrink': median MSE ratio vs raw	0.941 (19)	0.941 (19)
vibration	recalibration 'affine': median MSE ratio vs raw	0.962 (20)	0.962 (20)
vibration	recalibration 'anchored': median MSE ratio vs raw	0.709 (27)	0.709 (27)
temperature	trained models beating best naive in a majority of runs	1.000	1.000
temperature	Ridge-Univariate vs TiDE: median MSE ratio	0.872 (18)	0.872 (18)
temperature	Ridge-Univariate vs median trained model: median MSE ratio	0.262 (18)	0.265 (18)
temperature	Ridge-Sensors vs TiDE: median MSE ratio	0.910 (13)	0.910 (13)
temperature	Ridge-Sensors vs median trained model: median MSE ratio	0.275 (18)	0.275 (18)
temperature	AR-Global vs TiDE: median MSE ratio	1.123 (5)	1.123 (5)
temperature	AR-Global vs median trained model: median MSE ratio	0.342 (18)	0.345 (18)
temperature	AR-Local vs TiDE: median MSE ratio	1.095 (5)	1.095 (5)
temperature	AR-Local vs median trained model: median MSE ratio	0.330 (18)	0.342 (18)
temperature	ChronosBolt-Small vs TiDE: median MSE ratio	0.973 (11)	0.973 (11)
temperature	ChronosBolt-Small vs median trained model: median MSE ratio	0.299 (18)	0.299 (18)
temperature	Chronos2 vs TiDE: median MSE ratio	0.954 (11)	0.954 (11)
temperature	Chronos2 vs median trained model: median MSE ratio	0.281 (18)	0.282 (18)
temperature	Ensemble-Mean-13 vs TiDE: median MSE ratio	1.790 (0)	1.790 (0)
temperature	Ensemble-Mean-13 vs median trained model: median MSE ratio	0.549 (17)	0.566 (17)
temperature	Ensemble-Median-13 vs TiDE: median MSE ratio	1.553 (0)	1.553 (0)
temperature	Ensemble-Median-13 vs median trained model: median MSE ratio	0.481 (18)	0.482 (18)
temperature	best naive (reference)	0.330	0.330
temperature	recalibration 'shrink': median MSE ratio vs raw	1.043 (0)	1.036 (1)
temperature	recalibration 'affine': median MSE ratio vs raw	1.042 (0)	1.040 (0)
temperature	recalibration 'anchored': median MSE ratio vs raw	0.399 (18)	0.449 (18)

Table S17: Hyperparameter check: every method on the 18 tuning runs. Oracle rows select on the test block and are shown for reference only.

Kind	Method	Median MSE	Ratio to TiDE orig.	Beats naive
temperature	Ridge-Univariate	0.2130	0.785	9/9
temperature	Ridge-Sensors	0.2651	0.842	9/9
temperature	TiDE-Oracle (ref.)	0.2263	0.895	9/9
temperature	TiDE-Tuned	0.2263	0.896	9/9
temperature	ChronosBolt-Small	0.2943	0.906	9/9
temperature	Chronos2	0.2660	0.921	8/9
temperature	Tuned-Any	0.2263	0.963	9/9
temperature	TiDE-cfg0-rerun	0.2577	1.000	9/9
temperature	TiDE-Original	0.2582	1.000	9/9
temperature	TimeXer-Oracle (ref.)	0.2616	1.013	9/9
temperature	TimeXer-Tuned	0.2622	1.035	9/9
temperature	AR-Local	0.3975	1.282	8/9
temperature	Best-Naive	0.3825	1.306	0/9
temperature	TimeXer-cfg0-rerun	0.3718	1.347	5/9
temperature	TimeXer-Original	0.3589	1.355	6/9
temperature	NHITS-Oracle (ref.)	0.4271	1.636	2/9
temperature	NHITS-cfg0-rerun	0.4691	1.717	0/9
temperature	NHITS-Original	0.4947	1.916	0/9
temperature	NHITS-Tuned	0.4332	2.142	0/9
vibration	TiDE-Oracle (ref.)	0.0088	0.994	9/9
vibration	TiDE-Original	0.0090	1.000	9/9
vibration	TiDE-cfg0-rerun	0.0090	1.000	9/9
vibration	TiDE-Tuned	0.0090	1.004	9/9
vibration	TimeXer-Oracle (ref.)	0.0087	1.027	9/9
vibration	Tuned-Any	0.0089	1.028	9/9
vibration	Ridge-Univariate	0.0090	1.035	9/9
vibration	Ridge-Sensors	0.0096	1.049	9/9
vibration	TimeXer-Tuned	0.0089	1.052	9/9
vibration	AR-Local	0.0093	1.086	9/9
vibration	Chronos2	0.0096	1.093	9/9
vibration	ChronosBolt-Small	0.0097	1.094	9/9
vibration	TimeXer-cfg0-rerun	0.0106	1.155	9/9
vibration	TimeXer-Original	0.0103	1.166	8/9
vibration	NHITS-Oracle (ref.)	0.0124	1.277	6/9
vibration	Best-Naive	0.0109	1.290	0/9
vibration	NHITS-Tuned	0.0141	1.429	1/9
vibration	NHITS-Original	0.0149	1.468	1/9
vibration	NHITS-cfg0-rerun	0.0158	1.567	0/9

Table S18: Hyperparameter check: all pre-registered paired tests.

Kind	A	B	Ratio A/B	A better	p	p_{Holm}
vibration	TiDE-Tuned	TiDE-Original	1.004	2/9	0.16	0.98
vibration	TimeXer-Tuned	TimeXer-Original	0.878	9/9	3.9×10^{-3}	0.04
vibration	NHITS-Tuned	NHITS-Original	0.948	8/9	0.05	0.38
vibration	TiDE-Tuned	TiDE-cfg0-rerun	1.004	2/9	0.20	1.00
vibration	TimeXer-Tuned	TimeXer-cfg0-rerun	0.911	8/9	0.03	0.22
vibration	NHITS-Tuned	NHITS-cfg0-rerun	0.941	5/9	0.30	1.00
vibration	Tuned-Any	TiDE-Original	1.028	3/9	0.43	1.00
vibration	Tuned-Any	Best-Naive	0.779	9/9	3.9×10^{-3}	0.04
vibration	Tuned-Any	Ridge-Univariate	0.996	7/9	0.50	1.00
vibration	Tuned-Any	Ridge-Sensors	0.964	6/9	0.43	1.00
vibration	Tuned-Any	Chronos2	0.933	8/9	0.01	0.11
temperature	TiDE-Tuned	TiDE-Original	0.896	9/9	3.9×10^{-3}	0.04
temperature	TimeXer-Tuned	TimeXer-Original	0.883	9/9	3.9×10^{-3}	0.04
temperature	NHITS-Tuned	NHITS-Original	0.999	5/9	0.65	1.00
temperature	TiDE-Tuned	TiDE-cfg0-rerun	0.896	9/9	3.9×10^{-3}	0.04
temperature	TimeXer-Tuned	TimeXer-cfg0-rerun	0.841	9/9	3.9×10^{-3}	0.04
temperature	NHITS-Tuned	NHITS-cfg0-rerun	1.040	3/9	0.30	0.90
temperature	Tuned-Any	TiDE-Original	0.963	7/9	0.13	0.52
temperature	Tuned-Any	Best-Naive	0.755	9/9	3.9×10^{-3}	0.04
temperature	Tuned-Any	Ridge-Univariate	1.226	0/9	3.9×10^{-3}	0.04
temperature	Tuned-Any	Ridge-Sensors	1.160	1/9	0.04	0.20
temperature	Tuned-Any	Chronos2	1.044	4/9	0.65	1.00

Table S19: Configuration selection on validation blocks, and reproducibility of the original configuration when re-trained on different hardware (test MSE ratio per run-fold).

Kind	Model	Folds	Orig. chosen (%)	= oracle (%)	Re-run/stored min/med/max
temperature	NHITS	45	11	13	0.00 / 0.97 / 1.82
temperature	TiDE	45	0	64	0.96 / 1.00 / 1.01
temperature	TimeXer	45	9	44	0.67 / 1.02 / 1.93
vibration	NHITS	45	13	20	0.44 / 1.01 / 1.30
vibration	TiDE	45	7	13	0.98 / 1.00 / 1.02
vibration	TimeXer	45	7	42	0.75 / 1.01 / 1.16

Table S20: Training-seed variability (seeds 42, 101, 202; 14 iteration-1 runs).

Kind	Model	Runs	Median CV	Median max/min	Worst max/min	Seed 42 best (%)
temperature	TiDE	6	0.010	1.020	1.04	33
temperature	MLP	6	0.032	1.064	1.15	33
temperature	KAN	6	0.040	1.082	1.74	33
temperature	DeepNPTS	6	0.050	1.101	1.19	0
temperature	TimeXer	6	0.051	1.103	1.26	50
temperature	NBEATSx	6	0.054	1.107	1.46	17
temperature	DilatedRNN	6	0.064	1.136	1.45	50
temperature	ALL 13	6	0.070	1.145	2.83	37
temperature	NHITS	6	0.088	1.168	1.29	50
temperature	BiTCN	6	0.100	1.188	1.35	17
temperature	TFT	6	0.137	1.305	1.51	50
temperature	TSMixerx	6	0.150	1.309	1.43	50
temperature	TCN	6	0.250	1.555	1.98	50
temperature	TabNet	6	0.340	2.014	2.83	50
vibration	TiDE	8	0.003	1.007	1.01	62
vibration	DeepNPTS	8	0.017	1.035	1.10	75
vibration	TimeXer	8	0.017	1.033	1.12	38
vibration	NBEATSx	8	0.023	1.043	1.19	0
vibration	TCN	8	0.026	1.052	1.10	38
vibration	MLP	8	0.026	1.053	1.14	38
vibration	ALL 13	8	0.031	1.058	3.05	38
vibration	NHITS	8	0.034	1.066	1.44	50
vibration	DilatedRNN	8	0.034	1.068	1.15	12
vibration	BiTCN	8	0.035	1.067	1.12	50
vibration	TFT	8	0.042	1.084	1.12	38
vibration	TabNet	8	0.044	1.089	3.05	38
vibration	TSMixerx	8	0.046	1.090	1.14	38
vibration	KAN	8	0.060	1.125	2.06	25

Table S21: Headline conclusions recomputed at each seed and for the per-model mean over seeds.

Kind	Seed	$N_>$	Ridge/TiDE	Ridge/median	TiDE best	Anchored
vibration	42	2	1.031 (0/8)	0.601	8/8	0.746
vibration	101	2	1.024 (0/8)	0.605	8/8	0.753
vibration	202	2	1.034 (0/8)	0.595	8/8	0.738
vibration	seed-mean	3	1.032 (0/8)	0.671	8/8	0.801
temperature	42	2	0.840 (6/6)	0.235	5/6	0.422
temperature	101	2	0.824 (6/6)	0.258	6/6	0.438
temperature	202	1	0.823 (6/6)	0.238	6/6	0.416
temperature	seed-mean	2	0.834 (6/6)	0.303	5/6	0.473

Table S22: Zero-shot cross-domain transfer (fold F5, 80 origins per cell). Mean MSE is in standardised units of the source domain.

Channel	Model	Shift	Cells	Mean MSE	Median ratio to in-domain
self_x_ac	TiDE	both	36	0.0413	1.001
self_x_ac	TiDE	diagonal	9	0.0389	1.000
self_x_ac	TiDE	spatial-only	18	0.0377	1.001
self_x_ac	TiDE	temporal-only	18	0.0401	1.012
self_x_ac	TimeXer	both	36	0.0453	1.018
self_x_ac	TimeXer	diagonal	9	0.0431	1.000

Channel	Model	Shift	Cells	Mean MSE	Median ratio to in-domain
self_x_ac	TimeXer	spatial-only	18	0.0464	1.001
self_x_ac	TimeXer	temporal-only	18	0.0446	1.010
self_x_ac	TiDE	temporal (backward)	27	0.0173	1.014
self_x_ac	TiDE	temporal (forward)	27	0.0646	1.000
self_x_ac	TimeXer	temporal (backward)	27	0.0190	1.031
self_x_ac	TimeXer	temporal (forward)	27	0.0712	1.006
self_temperature_one	TiDE	both	36	0.2289	1.046
self_temperature_one	TiDE	diagonal	9	0.2099	1.000
self_temperature_one	TiDE	spatial-only	18	0.2304	1.084
self_temperature_one	TiDE	temporal-only	18	0.2093	1.045
self_temperature_one	TimeXer	both	36	0.2374	1.035
self_temperature_one	TimeXer	diagonal	9	0.2753	1.000
self_temperature_one	TimeXer	spatial-only	18	0.2525	1.111
self_temperature_one	TimeXer	temporal-only	18	0.2527	1.055
self_temperature_one	TiDE	temporal (backward)	27	0.1827	1.087
self_temperature_one	TiDE	temporal (forward)	27	0.2620	1.021
self_temperature_one	TimeXer	temporal (backward)	27	0.2066	1.114
self_temperature_one	TimeXer	temporal (forward)	27	0.2783	0.991

Table S23: Variance-floored Gaussian likelihood training (iteration 1, unit = target). The floor was derived from the full record and is therefore not a causal method; reported for completeness.

Model	Targets	Median change (%)	Improved	p
TSMixerx	26	-2.83	18/26	0.15
TiDE	26	+0.16	9/26	0.71

Table S24: Pilot run that fixed the roster (bearing temperature, iteration 1, earlier overlapping split scheme): MSE divided by persistence MSE. FEDformer produced no finite forecasts.

Model	R1	R2	R3	Mean
TiDE	0.86	0.90	0.43	0.73
TimeXer	1.01	1.29	0.60	0.97
Naive-Persistence	1.00	1.00	1.00	1.00
NHITS	2.07	1.54	1.01	1.54
TFT	2.20	2.41	1.19	1.93
BiTCN	2.82	3.31	1.59	2.57
TSMixerx	4.16	4.29	2.37	3.61
GRU	5.04	7.50	5.45	6.00
LSTM	5.75	7.01	5.42	6.06
TabNet	6.87	7.70	3.63	6.07
Naive-MA60	4.96	12.89	11.82	9.89
FEDformer	–	–	–	–

Table S25: Final automated verification run (script `a06_final_audit.py`).

Automated check	Result
privacy: no identifiers in any file or file name	pass
a01 rebuild identical	pass
a02 no FAIL verdicts	pass
a03 all 45 runs pass	pass
independent recomputation matches tables (8 random runs)	pass
b09 tuning integrity: 90/90 run-folds, origins & truth identical, validation before test	pass
tuning runs present (18) with account record	pass
tuning selection recomputed independently (5 random runs): same configuration, same test MSE	pass
worst diverged cell recomputed from raw arrays	pass
b10 extra-seed runs: 28 present, origins & truth identical to seed 42, no NaN	pass
seed comparison recomputed independently (vibration, Ridge-Univariate vs TiDE, each seed)	pass
report text present & matches table: 'ratio 0.94 on vibration'	pass
report text present & matches table: '(19/27 runs better, p = 0.06'	pass
report text present & matches table: '**worsens temperature in\n 18/18 runs** (ratio 1.04)'	pass
report text present & matches table: 'Spearman -0.27, p = 0.17'	pass
report text present & matches table: '(-0.00, p = 0.99)'	pass
report text present & matches table: 'temperature 1.000-1.004'	pass

Automated check	Result
report text present & matches table: 'vibration 0.67-0.98'	pass
report text present & matches table: '(27/27 vibration and 17-18/18 temperature'	pass
report text present & matches table: 'TSMixerx -2.8% median'	pass
report text present & matches table: '(18/26 targets better, p = 0.15)'	pass
report text present & matches table: 'TiDE +0.2% (p = 0.71)'	pass
report text present & matches table: '(34,248 vs 63,461)'	pass
report text present & matches table: 'R2 is -0.18 to 0.49 on vibration (median 0.04)'	pass
report text present & matches table: '0.91 to 0.99 on\n temperature (median 0.955)'	pass
report text present & matches table: 'costs 46% of Region 3'	pass
report text present & matches table: 'In 41 of 2,925'	pass
report text present & matches table: '(10 cells above 20x, 2 above 50x)'	pass
report text present & matches table: 'fold F2: 526x'	pass
report text present & matches table: '(median ratio 0.399 -> 0.449, still 18/18 runs)'	pass
report text present & matches table: '(1.243x / 1.553x TiDE'	pass
report text present & matches table: '0.96-1.02 times the stored value'	pass
report text present & matches table: '0.67-1.93 for TimeXer and 0.00-1.82 for NHITS'	pass
report text present & matches table: '2 of 2,160 trainings failed'	pass
report text present & matches table: '(median MSE ratio 0.896, 9/9 runs)'	pass
report text present & matches table: '(0.883, 9/9)'	pass
report text present & matches table: '(tuned TiDE 1.147x ridge, 0/9)'	pass
report text present & matches table: 'test-selected TiDE configuration is\n 1.147x ridge'	pass
report text present & matches table: '(1.004, better in 2/9 runs)'	pass
report text present & matches table: '(0.996, 7/9 runs,\n p = 0.50)'	pass
report text present & matches table: '(+0.40, p = 0.040)'	pass
report text present & matches table: '(-0.42, p = 0.031)'	pass
report text present & matches table: '(both p = 0.12)'	pass
report text present & matches table: '(all four p >= 0.27)'	pass
report text present & matches table: '28 complete'	pass
report text present & matches table: 'best of the three seeds in 38% (vibration) and\n 37% (temperature)'	pass
report text present & matches table: '1.024-1.034x TiDE'	pass
report text present & matches table: '(0.823-0.840x)'	pass
report text present & matches table: 'improves every\n run at every seed'	pass
report text present & matches table: '(median coefficient of\n variation 0.003 / 0.010)'	pass
report text present & matches table: 'TiDE alone gets 1.008 (vibration, improved in 7/27 runs) and\n 1.006 (temperature, 7/18)'	pass
report text present & matches table: '1.146x ridge there (better in 3/18 runs)'	pass
report text present & matches table: 'trained model ahead of ridge (21/27 runs)'	pass
report text present & matches table: '(LSTM 6.06x, GRU 6.00x, TabNet 6.07x persistence MSE; FEDformer NaN)'	pass
report text present & matches table: 'ratios range\n from 3.6x to 7.7x'	pass
report text present & matches table: 'TabNet did as badly in the\n pilot (6.07x persistence) as LSTM (6.06x) and GRU (6.00x)'	pass
report text present & matches table: 'TabNet (0.340) and TCN (0.250)'	pass
key_numbers.json present	pass

Table S26: Settings of the 13 trained models, fixed once without tuning. Shared settings (NeuralForecast models): input 128 min, 15-step output, learning rate 10^{-3} (Adam), batch size 32, window batch 256 where applicable, at most 3000 steps, validation every 50 steps, early stopping after 15 checks without improvement, gradient clipping 1.0, per-window robust scaler, seed 42, all non-target channels as historical covariates.

Model	Architecture settings
NHITS	library defaults (three MLP stacks)
TiDE	hidden size 256; other settings library defaults
BiTCN	hidden size 32
TFT	hidden size 64, 4 heads, dropout 0.1, window batch 64
TSMixerx	2 blocks, feed-forward size 64
TimeXer	hidden size 128, 4 heads, 2 encoder layers, patch length 16
NBEATSx	library defaults
MLP	hidden size 256, 3 layers
TCN	encoder and decoder hidden size 128
DilatedRNN	encoder hidden size 128
KAN	library defaults
DeepNPTS	hidden size 128
TabNet	pytorch-tabnet regressor on the flattened 128-minute history; $n_d = n_a = 32$, 4 steps, $\gamma = 1.5$, batch 512 (virtual 128), at most 100 epochs, patience 15

Table S27: Hyperparameter check: configuration 0 is the original setting; configurations 1–7 were drawn without replacement from the grid with random seed 2026 before any run.

Model	Original (config. 0)	Grid
TiDE	256 / 0.3 / 10^{-3} / 1 layer	hidden {128, 256, 512}; dropout {0.1, 0.3}; learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$; encoder layers {1, 2}
NHITS	512 / 0.0 / 10^{-3}	MLP width {256, 512, 1024}; dropout on θ {0.0, 0.2}; learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$
TimeXer	128 / 0.1 / 10^{-3}	hidden {64, 128, 256}; dropout {0.1, 0.3}; learning rate $\{3\times10^{-4}, 10^{-3}, 3\times10^{-3}\}$

Table S28: Columns of the released files (56 feature columns plus identifiers). Prefixes: self = reference motor, nb1/nb2 = neighbours. All rolling quantities are trailing and computed within segments. Region 3 has no second neighbour; its nb2 columns are zero.

Column	Meaning
`unique_id`, `ds`	series identifier and UTC timestamp (1-min)
`segment_id`	index of the gap-free segment
`y`	copy of `self_temperature_one` (the original target column)
`y_roll_mean_15`	trailing 15-min mean of y
`y_roll_std_15`	trailing 15-min standard deviation of y
`y_roll_std_60`	trailing 60-min standard deviation of y
`y_roll_std_240`	trailing 240-min standard deviation of y
`y_diff_1`	1-min difference of y
`y_diff_5`	5-min difference of y
`y_diff_15`	15-min difference of y
`y_diff_60`	60-min difference of y
`y_roll_iqr_60`	trailing 60-min inter-quartile range of y
`self_temperature_one`	self: drive-end bearing temperature (°C, PT100, corrected)
`self_temperature_two`	self: motor-body temperature (°C)
`self_t_grad`	self: bearing minus body temperature
`self_x_ac`	self: x-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`self_x_ac_mean_60`	self: trailing 60-min mean of the x-axis amplitude
`self_x_ac_std_60`	self: trailing 60-min standard deviation of the x-axis amplitude
`self_y_ac`	self: y-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`self_y_ac_mean_60`	self: trailing 60-min mean of the y-axis amplitude
`self_y_ac_std_60`	self: trailing 60-min standard deviation of the y-axis amplitude
`self_z_ac`	self: z-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`self_z_ac_mean_60`	self: trailing 60-min mean of the z-axis amplitude
`self_z_ac_std_60`	self: trailing 60-min standard deviation of the z-axis amplitude
`self_vib_ac`	self: vibration magnitude: Euclidean norm of the three axis amplitudes (m/s^2)
`self_vib_dc`	self: static acceleration: norm of the per-minute mean of each axis (m/s^2, $\approx g$)
`nb1_temperature_one`	nb1: drive-end bearing temperature (°C, PT100, corrected)
`nb1_temperature_two`	nb1: motor-body temperature (°C)
`nb1_t_grad`	nb1: bearing minus body temperature
`nb1_t1_diff_15`	nb1: 15-min change of bearing temperature
`nb1_x_ac`	nb1: x-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`nb1_x_ac_mean_60`	nb1: trailing 60-min mean of the x-axis amplitude
`nb1_x_ac_std_60`	nb1: trailing 60-min standard deviation of the x-axis amplitude
`nb1_y_ac`	nb1: y-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`nb1_y_ac_mean_60`	nb1: trailing 60-min mean of the y-axis amplitude
`nb1_y_ac_std_60`	nb1: trailing 60-min standard deviation of the y-axis amplitude
`nb1_z_ac`	nb1: z-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`nb1_z_ac_mean_60`	nb1: trailing 60-min mean of the z-axis amplitude
`nb1_z_ac_std_60`	nb1: trailing 60-min standard deviation of the z-axis amplitude
`nb1_vib_dc`	nb1: static acceleration: norm of the per-minute mean of each axis (m/s^2, $\approx g$)
`nb2_temperature_one`	nb2: drive-end bearing temperature (°C, PT100, corrected)
`nb2_temperature_two`	nb2: motor-body temperature (°C)
`nb2_t_grad`	nb2: bearing minus body temperature
`nb2_t1_diff_15`	nb2: 15-min change of bearing temperature
`nb2_x_ac`	nb2: x-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
`nb2_x_ac_mean_60`	nb2: trailing 60-min mean of the x-axis amplitude

Column	Meaning
nb2_x_ac_std_60	nb2: trailing 60-min standard deviation of the x-axis amplitude
nb2_y_ac	nb2: y-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
nb2_y_ac_mean_60	nb2: trailing 60-min mean of the y-axis amplitude
nb2_y_ac_std_60	nb2: trailing 60-min standard deviation of the y-axis amplitude
nb2_z_ac	nb2: z-axis vibration amplitude: trailing 3-min standard deviation of raw samples (m/s^2)
nb2_z_ac_mean_60	nb2: trailing 60-min mean of the z-axis amplitude
nb2_z_ac_std_60	nb2: trailing 60-min standard deviation of the z-axis amplitude
nb2_vib_ac	nb2: vibration magnitude: Euclidean norm of the three axis amplitudes (m/s^2)
nb2_vib_dc	nb2: static acceleration: norm of the per-minute mean of each axis (m/s^2, $\approx g$)
d_self_nb2	self minus nb2 bearing temperature
d_self_nb1	self minus nb1 bearing temperature
tod_sin	time of day, sine
tod_cos	time of day, cosine